

Rakesh Kumar

Manager Experience Engineer at Publicis Sapiant

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SUMMARY

I hold the position of Manager of Experience Engineering at Publicis Sapiant and have over 16+ years of extensive experience in the IT industry. My expertise lies in designing solutions that adhere to best practices, compliance, and future perspectives. I have a proven track record of success using my skills in JavaScript, React, Node, Angular, microservices, microfrontend, performance, accessibility, and security to create user-friendly interfaces optimized for performance and security.

Highlights

- **Frontend:** Experienced in crafting and building various types of applications, including Single Page Applications (SPA) and those utilizing Server-Side Rendering (SSR), Static Site Generation (SSG), and Incremental Static Regeneration (ISR) architectures. Proficient in employing different frameworks such as Next.js (React) and Angular for these purposes. Expertise in Micro Frontend Architecture, Atomic Design principles, integrating REST services, converting PSD designs into HTML, and following BEM. Ensures responsive web design with compatibility across various browsers and platforms.
- **Backend:** Specializes in the design and development of RESTful services using Node.js, and is well-versed in Microservice Architecture. Proficiency extends to Python with Flask and Java with Spring Boot. Maintains comprehensive documentation for backend services.
- **Others:** Proficient in publishing npm packages, understanding monorepo principles, setting up and maintaining CI/CD pipelines, implementing diverse caching strategies, overseeing team management, and providing training to elevate their skills. Experienced in gathering business requirements and translating them into technical stories.
- **Blogger:** In addition to my full-time role, I am a dedicated part-time blogger, regularly writing and publishing technical blogs on Medium. You can find my work at <https://medium.com/@rakeshkumar-42819>

WORK EXPERIENCE

01 SAPIENT GLOBAL MARKET (Manager Experience Engineer) - *(Since March 2015 to Present)*

Invesco (ESG) as Frontend Architect

In my role as a Frontend Architect at Invesco, I was tasked with identifying and implementing solutions for an existing ESG (Environmental, Social, and Governance) application. This React-based application, developed several years ago, faced multiple challenges.

Key Issues:

- Inefficient user journey to access desired information
- Suboptimal page load performance
- Slow backend API response times
- Lack of code consistency, adherence to industry best practices, and overall code quality
- Absence of unit tests and end-to-end (e2e) testing

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Implemented Solution: To address these challenges, I implemented the following strategies:

- Comprehensive redesign: I overhauled not just the codebase, but also streamlined the user flow for improved navigation and accessibility.
- Performance optimization: By implementing code splitting, optimizing Core Web Vitals, and leveraging caching techniques, I reduced page load times from 60 seconds to 15 seconds.
- Backend collaboration: I initiated and participated in brainstorming sessions with the backend team to identify API bottlenecks and develop targeted solutions.
- Code quality enhancement: I introduced and enforced coding best practices to ensure consistency and quality. This involved integrating tools such as linters, Prettier for code formatting, commitlint for standardized commit messages, and Husky for pre-commit hooks.
- Testing implementation: I developed a comprehensive testing strategy, achieving 90% unit test coverage and implementing end-to-end testing using Playwright.

US Bank as Frontend Architect

As a Frontend Architect at US Bank, I was responsible for designing solutions for two public-facing applications. These projects required a complete redesign and rebuild to address several critical challenges:

Key Challenges:

- Generation and delivery of 8,000 pages across multiple locales
- Real-time synchronization with backend data to ensure up-to-date information daily
- Compliance with specific business requirements for SEO, accessibility, and performance
- Implementation of robust security measures for public-facing applications
- Enhancement of user experience through integration of Google Maps for advisor and branch searches

Implemented Solution: To address these challenges, I developed and implemented the following strategies:

- Architected a solution using NextJS framework, leveraging Incremental Static Regeneration (ISR) and Server-Side Rendering (SSR) for efficient page generation.
- Implemented the Atomic design pattern for modular and scalable component management.
- Developed a library of reusable components, publishing them to the US Bank's internal npm registry to facilitate use across multiple applications.
- Established a CDN-level caching system to serve static pages efficiently until cache expiration.
- Implemented critical security headers to protect against XSS, DDoS, and CSRF attacks.
- Designed and implemented an HMAC-based system for generating secure API authentication tokens.
- Created a separate micro-application for the Search functionality, enhancing the Google Maps integration while maintaining seamless interoperability with both main applications.
- Integrated robust localization support into both applications.
- Achieved 85% unit test coverage across all applications and implemented end-to-end testing using Playwright.

Invesco as Team Lead

Invesco is an independent investment management firm dedicated to delivering an investment experience that helps people get more out of life. They offer a wide range of single-country, regional and global capabilities across major equity, fixed income, and alternative asset classes, delivered through a diverse set of investment vehicles. High-quality results begin with specialized insight and disciplined oversight.

1. CWp (Collaboration Workplace) is the platform that facilitates collaboration, workflow management, and automation across team and organizations. So far, the platform is being used by Fund Accounting, connecting teams at Invesco and across third parties (State Street and BNYM). CWp based on Appian, used as fully customizable case management tool with advanced integration capabilities, and Symphony which is cloud-based collaboration tool with modern chat bot features.

As a part of ANG project, the plan is to extend CWp platform to other departments to facilitate oversight process and implement a future state workflow for issues resolution and collaboration.

Responsibilities:

- Gather information of each workflow need to be designed for every newly introduced department and creating high level flow diagrams.
- Following multi-tier architecture (Symphony Chat Bot, Extension App for Symphony using Next JS, Java Spring Boot APIs, MS SQL).
- Developed Symphony Chat bot with extension app which build using Next JS framework of React JS. To improve the performance and quickly load of view used prefetching to load the routes. Using SSG and SSR both strategies to render page as per need basis and as per user interaction.
- Also creating APIs to fetch data from various sources (MS SQL and Other third-party services).
- Implemented NLP (Natural Language Process) on Symphony side so users can create cases without spending much time to filling forms manual.
- Writing unit test cases on backend as well as on frontend side to make sure the code quality and reduce bugs.
- Following the agile environment which involved sprint planning and daily stand-up meeting that were managed using Jira.
- Also participating other ceremonies like sprint planning/grooming, backlog refinements, sprint review/demo and retrospectives.
- Used bit bucket for version control and main repository management and bamboo CI/CD pipeline for deployment.

Skills: Next JS/React JS, React Hooks, Context API, Java Spring Boot, Symphony Chat Bot, Typescript, JavaScript, Junit, Jest, Bamboo, Bitbucket

2. ExPRT (Expense Payment & Reporting System) allow Fund Financial Services to automate the fund expense payment, distribution, and reporting process, along with monthly fund expense analysis, management fee and hypothetical expense calculation. This application was build using Visual Basic and was not much user friendly. We propose client to rebuild this application as web application using latest tech stacks. Which makes huge impact on performance and user experience.

Responsibilities:

- Documented all the business logic and behavior of current running system.
- Responsible for creating front-end wireframes and prototypes using sketch and photoshop.
- Developed web-based application, frontend in Angular 7 and backend APIs using Java Spring Boot.
- Lead team of 4 members (2 Frontend Developer, 1 Java Developer, 1 QA) including writing APIs and implementing those APIs on frontend.
- Writing unit test cases on backend as well on frontend side to make sure the code quality and less chance of getting bugs.
- Following the agile environment which involved sprint planning and daily stand-up meeting that were managed using Jira.
- Used bit bucket for version control and main repository management and bamboo CI/CD pipeline for deployment.

Skills: Angular 7, Redux, Java Spring Boot, Typescript, JavaScript, Junit, Jasmine, Bamboo, Bitbucket, Sketch, Photoshop

Thesys CAT

The CAT is a Consolidated Audit Trail that tracks orders throughout their lifecycle and identifies the exchanges and broker-dealers handling them, thus allowing regulators to track activity more efficiently and accurately in eligible securities — those under the jurisdiction of the Securities and Exchange Commission (the "SEC") — throughout the U.S. markets. The CAT is created by a joint plan (CAT NMS Plan) of the Plan Participants or simply "Participants."

Responsibilities as Full Stack Developer:

- Work closely with Sr. Architect to finalise the architecture of project and creating high level architecture diagrams.
- Followed multi-tier architecture for overall project but on backend followed microservice architecture.
- Responsible to manage frontend in Angular 5 and backend APIs developed in Node (API Gateway), Python (microservices) and PostgreSQL.
- Leading team with 1 testing engineer and 1 frontend developer and resolving their dependencies if any.
- Expose APIs to use of frontend developers to display data on UI
- Writing unit test cases on backend and e2e testing frontend side to make sure the code quality and reduce bugs.

- Used git for version control as main repository management, AWS and Jenkins for CI/CD pipeline.
- Following the agile environment which involved sprint planning and daily stand-up meeting that were managed using Jira.
- Also participating other ceremonies like sprint planning/grooming, backlog refinements, sprint review/demo and retrospectives.

Skills: JavaScript, ES6, NodeJS, Angular 5, Express, Python with Flask, Jasmine, Pytest, lodash, ESLint, GIT, VIM, Jenkins

Lloyds

Lloyds Bank plc is a British retail and commercial bank with branches across England and Wales. It has traditionally been considered one of the "Big Four" clearing banks. We need to develop and upgrade the whole banking and transaction process securely.

Responsibilities as Sr. Node JS Developer:

- Followed microservice architecture and distributed whole banking servers in independent services (transaction, payment, banking, etc).
- Responsible to develop the APIs in NodeJS as per the different workflow and consuming APIs developed by other teams.
- Writing unit test cases on NodeJS side using jasmine and karma.
- Following the agile environment which involved sprint planning and daily stand-up meeting that were managed using Jira.
- Also participating other ceremonies like sprint planning/grooming, backlog refinements, sprint review/demo and retrospectives.

Skills: JavaScript, ES6, Node, Loopback, Jasmine, lodash, ESLint, GIT, Jenkins

Bank of Montreal (BMO - Client Connect)

The Bank of Montreal is a Canadian multinational investment bank and financial services company. Founded in Montreal, Quebec, in 1817 as Montreal Bank, its head office remains in Montreal, while its operational headquarters and executive offices have been located in Toronto, Ontario since 1977.

We need to develop the screen where user can see all the list of ETFs product list and validate how these ETF funds performing. We used NodeJS with server-side rendering as a solution and integrated the within BMO existing website without impacting their current running application.

Responsibilities as Sr. NodeJS Developer:

- Initial setup of project with NodeJS, Express JS and gulp as bundler.
- Used EJS (embedded JS) templating to render data on front-end.
- Created functionality of Download PDF and CSV for performance data and factsheet which contains the information of funds
- Implemented the High Chart with custom modification to show the performance of funds.

Skills: Node JS, Express JS, JavaScript, ES5, EJS, Jasmine, lodash, JSHint, HTML/CSS, RWD, jQuery, GIT

02 DAMCO GROUP (Sr. Software Engineer) - (Since March 2013 to March 2015)

Responsibilities included front-end development with HTML, CSS, JavaScript, WordPress, Angular JS and Graphic Designing of website, Hybrid and Native Apps. Used Bootstrap and custom CSS to make the websites responsive and browser compatible.

- Develop Angular App using IONIC framework
- Develop more than 10 websites used WordPress
- Used JavaScript, jQuery and CSS3 to make website more interactive.
- Used latest technologies and best practices and web standards.

Skills: HTML5, CSS2/3, jQuery, JavaScript, Bootstrap, Angular, IONIC, RWD, Cross Browser Compatible Websites, WordPress, Photoshop

03 18th TECHNOLOGY (Sr. Web Designer) - *(Since January 2011 to March 2013)*

Under this organization responsibility included front-end development of website, maintaining website, handling team of 5 members, creating SEO friendly websites, theme implementation for WordPress and designing the UI and logo of websites and mobile Apps.

Skills: HTML5, CSS2/3, jQuery, JavaScript, Bootstrap, RWD, Cross Browser Compatible Websites, WordPress, Photoshop, Illustrator

04 MAGNON SOLUTION (Assistant Manager) - *(Since June 2010 to January 2011)*

Responsibility included front-end development of website, maintaining website, handling team, creating SEO friendly websites, theme implementation for WordPress and designing the UI and logo of websites.

Skills: HTML5, CSS2/3, jQuery, JavaScript, Bootstrap, Responsive Web Designing, Cross Browser Compatible Websites, Photoshop, Illustrator

05 MOSAIC ITES SERVICES (Web Designer) - *(Since July 2008 to June 2010)*

Responsibility included front-end development of website, maintaining website, creating SEO friendly websites and designing the UI and logo of websites.

Skills: HTML, CSS, jQuery, JavaScript, Cross Browser Compatible Websites, Photoshop, Illustrator

QUALIFICATIONS

Graduation in B.A. Political Science Honors from Delhi University.

CERTIFICATIONS

01 ADIM (Advance Diploma in Multimedia)

2005 – 2008
Arena Multimedia - APTECH

02 CIC (Certificate in Computing)

2005
Indira Gandhi National Open University

03 MEAN Stack

June 2021
Lynda.com | License 5E519E

04 Symphony Certified Bot Developer

2005
Symphony | ID: E9D4-2D28-014E-4CD9

05 MEAN Stack

November 2020
Publicis Sapient

SKILLS & TOOLS

- JavaScript
- Node JS
- Express, Loopback
- React JS
- Next Js
- Angular Js
- Redux/API Context
- Basic of AWS
- Webpack
- SSG/SSR/CSR
- HTML/HTML5
- CSS/CSS3/BEM
- SCSS/LESS
- Responsive Web Designing
- CI/CD
- GIT
- UI/UX Designing
- Photoshop/Sketch/Figma
- Docker
- Kubernetes

PERSONAL INFO

Father's Name : Late. Shr. Ram Iqbal Das
Date of Birth : 16th Sep, 1984

Marital Status : Married
Language : Hindi, English

Nationality : Indian